

National Information Ecosystem Executive Brief

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Decision required. Treat AI readiness as an institutional and statistical-system question before treating it as a technology procurement question. Authorize a bounded ecosystem diagnostic, resolve the most consequential data and accountability gaps, and allow an AI pilot only when purpose, authority, evidence, safeguards and recurrent capacity are explicit.

Executive conclusion

Official statistics are produced through a national information ecosystem: public institutions, sector producers, administrative systems, research bodies, private data holders, technology providers, funders, oversight bodies and users. A capable national statistical office remains central, but it cannot guarantee timely and trustworthy information if definitions conflict, exchanges fail, identifiers do not align, source coverage is biased, experimental results are generalized too far, or operating capacity depends on temporary projects.

The appropriate leadership response is therefore not a general instruction to use AI. It is a controlled sequence: define the decision need; map actors, assets and flows; test fitness for use; establish authority and safeguards; improve non-AI dependencies; and then screen a narrow AI-assisted task against a credible baseline.

What leaders should authorize now

Action	Leadership purpose	Minimum evidence
1 Scope one decision chain	Keep the diagnostic tied to a public decision and deadline.	Named user, product, population, geography, reference period and consequence of error.
2 Name accountable roles	Separate source ownership, production, methods, coordination, technology and release authority.	Mandate references, responsible roles and unresolved authority questions.
3 Map critical assets and flows	Expose where meaning, quality, time or accountability is lost.	Stable identifiers, provenance, handoffs, transformations, controls, delays and fallback.
4 Resolve priority dependencies	Avoid using AI to conceal weak metadata, access or coordination.	Action owners, deadlines and verification evidence.
5 Gate any AI pilot	Protect official-statistics quality and public trust.	Baseline, benchmark, human review, security boundary, stop rules and funded operating model.

The institutional shift

An organization chart shows where units sit. An ecosystem map shows how information actually moves and where responsibility changes. One institution may own operational records, another may produce statistics, another may approve methods, and another may operate the technical platform. When these roles are collapsed into a single institutional label, accountability gaps and competing claims remain hidden.

Role	Executive question
Source owner	Who controls the original operational or observed data?
Statistical producer	Who transforms sources into a statistical output?
Methodological authority	Who approves concepts, methods, revisions and release?
Steward	Who maintains metadata, access rules, quality evidence and change communication?
Coordinator	Who resolves duplication and sets common standards?
Technology provider	Who operates infrastructure, models and support?
Oversight body	Who enforces legal, confidentiality, security and ethical requirements?
User or affected group	Who makes decisions or bears the consequences?

Evidence discipline

Every material claim should be marked as evidenced, reported or unknown. Unknown is a legitimate diagnostic result. It identifies the evidence owner and prevents informed opinion from becoming an institutional fact.

Priority risk picture

Risk	Why it matters	Immediate control
Conflicting concepts and revisions	Different estimates can both appear official while measuring different things.	Agree definitions, classifications, revision status and provenance.
Coverage and selection bias	Timely administrative, extension, geospatial or private sources may omit critical populations.	Measure coverage against the target population and validate by domain.
Missing identifiers	Linkage can introduce exclusions, false matches and untraceable transformations.	Establish governed linkage rules and quantify linkage error.
Narrow pilot evidence	High accuracy in accessible districts may not generalize nationally.	Use independent, geographically stratified validation and explicit

Risk	Why it matters	Immediate control
		information boundaries.
Fragile operational capacity	A pilot may depend on one analyst, short funding or an external supplier.	Fund recurrent roles, reproducible pipelines, handover and fallback.
Confidentiality and unlawful reuse	Improper inputs can breach law, trust and statistical confidentiality.	Confirm legal basis, minimization, access controls and disclosure review.

Kintaba decision example

The fictional Republic of Kintaba needs district crop-production information six weeks before its annual survey estimate is released. The ecosystem includes a national statistics directorate, agriculture ministry, extension service, land administration, meteorological service, food-security secretariat, research institute, university geospatial laboratory, market service, mobile operator, farmer organizations and government digital service.

The case deliberately contains contradictions: maize production is reported as both 4.8 and 4.2 million tonnes without visible definitions or revision status; extension coverage is stronger near roads; 19 of 82 weather stations have substantial missing observations; holdings, parcels, extension reports and survey units lack a common identifier; satellite accuracy is reported from 600 labelled fields in only three accessible districts; and the geospatial project has eight months of funding and one trained analyst.

Decision on the proposed automated estimate

A proposal to use generative AI and satellite imagery to produce district crop estimates automatically should not proceed as stated. The system lacks the evidence and operating controls required for an automated official estimate. A defensible next step is to reconcile concepts and revision status, document the early-estimate protocol, establish an independent benchmark, assess geographic and subgroup performance, confirm access and security, and name the human authority responsible for review and release.

A narrower controlled pilot may later test satellite classification for selected crops and districts, or assist analysts in documenting discrepancies and missing metadata. Generative AI should not act as the estimator, methodological authority or release authority.

AI pilot authorization gate

Gate	Question that must have an evidenced answer
Purpose	What exact decision and statistical product will improve?
Baseline	What current method, timing, cost and error evidence will the pilot be compared with?
Data	Which sources are permitted, representative, documented and operationally stable?

Gate	Question that must have an evidenced answer
Quality	What benchmark, domains, seasons, subgroups and tolerances are required?
Authority	Who owns the estimate, method, revision, review and release decision?
Human review	Which qualified official can challenge, override or stop the system?
Security	What information may enter each tool and environment?
Sustainability	What recurrent skills, funding, infrastructure, supplier and fallback arrangements exist?

Recommended ninety day leadership agenda

Period	Leadership action	Output
Days 1 to 15	Appoint sponsor, statistical lead, domain lead, stewardship lead and security contact. Select one decision chain.	Approved scope and accountable roles.
Days 16 to 40	Map actors, authority, assets, flows, users, quality evidence and unknowns.	Evidence-backed ecosystem map.
Days 41 to 60	Prioritize no more than five gaps by consequence and dependency.	Owned action register and verification plan.
Days 61 to 75	Define the baseline and screen one bounded AI-assisted task.	Proceed, investigate or do-not-pursue recommendation.
Days 76 to 90	If eligible, approve a controlled pilot protocol with stop rules and independent review.	Signed pilot charter or documented decision not to pilot.

Measures of progress

- Critical actors and roles have stable identifiers and documented authority.
- Priority data assets have owners, provenance, quality evidence, access conditions and known changes.
- One critical flow is traceable from source to user decision, including transformations and delays.
- Material claims are labelled as evidenced, reported or unknown.
- The leading non-AI dependencies have owners, deadlines and verification evidence.
- Any AI pilot has a baseline, independent benchmark, human authority, security boundary, stop rules and recurrent resources.

Leadership position

The responsible position is neither to reject AI categorically nor to authorize it generically. Leadership should create a governed route from public purpose to evidence, from evidence to controlled testing, and from testing to an accountable decision. In official statistics, the central question is not whether an algorithm can produce an output. It is whether the institution can explain what the output means, whom it represents, how uncertainty and error were assessed, who is responsible, and why the result is fit for the intended public use.

Proposed executive decision

Approve a ninety-day national information ecosystem diagnostic for one priority statistical decision chain. Do not authorize automated official estimates or operational AI deployment through this decision. Require the diagnostic team to return with an evidence-backed map, a prioritized dependency register, a baseline, and a recommendation to proceed, investigate further or stop.

Decision field	Record
Accountable sponsor	Name and position
Priority decision chain	Statistical product, user decision and deadline
Statistical lead	Name and position
Domain and stewardship leads	Names and positions
Security and legal contacts	Names and positions
Current AI authorization	Diagnostic and screening only; no automated release authority
Review date	Date for executive decision on any controlled pilot